The Value of Shared Mobility Data for Urban Traffic Monitoring

Working manuscript — author information to be supplied

***Abstract*—Can bike-sharing and car-sharing records support traffic monitoring when counts are unavailable? We distinguish estimation at new locations with no count history from reconstruction at known locations after a simulated sensor failure. Using Bologna data from 2023–2024, we train on 2023 counts and test spatially excluded locations in both years, and previously observed locations in 2024. We compare citywide sharing activity, nearby rental endpoints and straight-line origin–destination geometry against contextual baselines. The known-location baseline also includes frozen local traffic profiles from 2023; no contemporaneous counter measurements enter prediction. Descriptive maps reveal related urban polarities across services, but predictive benefits depend on the scenario and model. At new locations, sharing benefits are conditional, and pooled gains from citywide activity do not persist into 2024. At known bicycle locations, sharing-enhanced gradient boosting reduces pooled weighted absolute percentage error from 17.68% to 14.59%, improving all four sites. Poisson regression does not benefit, and the strongest known-location result for motor traffic remains the no-sharing baseline at 12.33%. The contribution is a scenario-specific assessment of sharing's incremental value: spatial transfer and reconstruction at known locations are distinct capabilities and must be evaluated separately.**

***Index Terms*—Bike-sharing, car-sharing, traffic reconstruction, sensor outage, spatial transfer, temporal validation.**

I. Introduction

TRAFFIC counts may be unavailable for two different reasons: a street section has never been monitored, or an existing counter has stopped working after accumulating a useful history. Sharing services can continue recording rentals in both situations. The practical question is whether these records improve estimates of general traffic under each information setting.

At a new location, a model must infer traffic from relationships learned elsewhere. At a known location, it can also use the historical traffic level and temporal profile. Poor spatial transfer therefore does not establish that sharing is unhelpful after a known sensor fails; successful reconstruction at a known location does not validate estimation at a new one. This distinction organises the study.

Bologna's bike-sharing and car-sharing services provide an informative comparison. Both record collection and return locations and times, but their users and operations differ [1], [2]. Bicycle rentals are mostly short intervals of use; car rentals can include stops and parking. We assess their predictive contribution using independent bicycle and motor-vehicle counts, without assuming that rental endpoints represent all passages through a street section.

Version: October 25, 2026. Bologna, 2023–2024.

Three research questions structure the paper:

1) *RQ1: urban rhythms.* Which temporal rhythms and geographical polarities recur across the two sharing services, and what do these observations measure?
2) *RQ2: new locations.* How much does sharing improve traffic estimation at locations whose counts have never been used for training, within the training year and in the following year?
3) *RQ3: known locations.* How much does sharing improve reconstruction in the following year when a location's previous counts are available, but its contemporaneous sensor measurements are not?

We address RQ2 by training on 2023 counts elsewhere and testing spatially excluded locations in 2023 and 2024. For RQ3, we train on the 2023 counts of all eligible locations and test the same locations in 2024. Neither experiment uses 2024 counts as inputs or for refitting. The pooled known-location model receives explicit local historical profiles, including in its no-sharing baseline. Within each experiment, matched comparisons test whether increasingly detailed sharing inputs improve estimates.

The contribution is an assessment of sharing's incremental value in two traffic-monitoring scenarios, alongside a description of urban rhythms. Relative to bicycle-volume extrapolation [3] and traffic-data reconstruction [4], [5], the focus is not a new estimator but the value of sharing under different information sets. Identical 2024 targets and scenario-specific baselines distinguish this value from the advantage of knowing a location's history. The comparison covers four bicycle locations and hundreds of motor-vehicle devices; it assesses sharing within each domain, not intrinsic predictability across transport modes.

Section II reviews related work. Section III describes the sharing data and Bologna's urban rhythms through temporal distributions, activity maps and rental balances. Section IV presents the sensor networks and experimental design. Section V reports and discusses pooled and local results for both scenarios, and Section VI concludes.

II. Related Work and Analytical Position

A. From Service Rhythms to Independent Traffic Counts

Bike-sharing reviews describe service-specific users, operating generations and usage patterns [1], [6]. Car-sharing demand depends on vehicle supply and operating conditions [2], and its spatial distribution relates to local context and alternative transport options [7]. Studies of nighttime activity and rental imbalances reveal temporal differences and recurring sources and sinks [8], [9]. This literature supports using endpoint records to describe urban activity, but does not establish that they represent general traffic.

RQ1 follows this descriptive approach by comparing the two services on matched geographical and temporal units. RQ2 and RQ3 then test their information against independent street-section counts. Rental balance measures neither street flow nor vehicle inventory, which also depends on relocation and maintenance. Agreement between sharing maps is therefore descriptive evidence, not independent validation of a traffic estimate.

B. Volume Extrapolation and the Value of Local Counts

Reviews and data-fusion studies frame bicycle-volume estimation as the combination of sparse counts with conventional and emerging sources [10]–[12]. Kaiser et al. [3] provide the closest precedent: they combine bike-sharing, crowd-sourced activity and contextual data in Berlin, validate at excluded counting locations, and investigate additional short-term counts at the predicted location. Their results demonstrate the importance of local supervision, not simply the availability of a geographically detailed activity source.

We adopt excluded-location validation but ask a narrower question: do citywide activity, nearby endpoints and straight-line origin–destination geometry improve an estimator with fixed contextual inputs? The cumulative input sets isolate the conditional contribution of each sharing representation, rather than benchmark multiple data sources. The known-location experiment also differs from adding sampled target-site dates across the observation period: an entire earlier year's history is available, and all evaluation periods follow it. Neither spatial validation nor the use of local counts is introduced here.

Graph-based bicycle estimation under sparsity [13] and Strava-based infrastructure assessment [14] further illustrate the range of inputs and monitoring purposes. For motor traffic, probe-vehicle volume estimation [15] supplies a related network-wide objective. These studies motivate testing spatial generalisation, but their trajectories, supervision and targets differ from rental endpoints. Absolute error comparisons across these settings would not identify a superior method.

C. Missing-Data Reconstruction and Available Information

Traffic imputation addresses incomplete observation networks. Lei et al. [5] develop Bayesian kernelized matrix factorisation for spatiotemporal imputation and kriging; Cini et al. [4] use graph message passing to reconstruct missing channels of multivariate time series. Reconstruction and spatial estimation are thus established methodological problems. Our two scenarios distinguish operational information settings, rather than introduce previously overlooked tasks.

The relevant distinction is which measurements remain available. Spatiotemporal reconstruction can exploit relationships among observed sensor streams. In our known-location experiment, no contemporaneous counter measurements enter prediction, including those from neighbouring locations. Only historical counts, contemporaneous sharing and context are available. The simulated missing interval also spans the full subsequent year, rather than isolated entries. Results for other missingness patterns or input configurations cannot be transferred without testing.

We therefore do not benchmark GRIN or matrix-completion methods. Instead, we test whether an external, selective mobility source adds value beyond frozen historical profiles. Providing the same local history to the no-sharing baseline prevents that history's contribution from being misattributed to sharing. Because sharing is observed during the period being reconstructed, this is not forecasting with unknown future predictors.

D. Endpoint Geometry Is Not Route or Demand Recovery

Observed bicycle route choice is not reducible to shortest distance [16], and preferences vary across cyclists and bicycle types [17]. A rental starting nearby may use another street, while through traffic can have distant endpoints. Car collection and return records also leave intermediate driving times unobserved. We therefore use the straight line between rental endpoints, termed a chord, as a spatial proxy, not a measured passage.

We do not reconstruct total origin–destination (OD) demand. Classical traffic-constrained inference [18], [19], Bayesian assignment approaches [20] and bicycle-specific OD procedures [21], [22] require additional inverse-modelling assumptions. Here OD geometry is an input and held-out section counts are the outcome. Accurate count estimates would not independently validate inferred routes or journeys.

E. Position and Contribution of This Study

The study combines independent traffic counts, separate new- and known-location experiments, and matched additions of sharing inputs. Both scenarios use the same 2024 outcomes, with scenario-specific baselines and pooled and local evaluation. Comparing bike-sharing and car-sharing within one city tests whether conclusions recur across services and observation networks. The design does not establish citywide accuracy, a controlled difference between transport modes or a universal ranking of data sources. It allows a positive known-location result to coexist with weak spatial transfer without conflating the two capabilities.

TABLE I
RENTAL COHORTS UNDER THEIR MODE-APPROPRIATE RULES. OUTSIDE-BOX RECORDS ARE EXCLUDED FROM THE STUDY, NOT NECESSARILY ERRONEOUS.

Year	Bike raw	Bike retained	Car raw	Car retained
2023	2,285,447	1,731,828	224,436	195,076
2024	3,106,337	2,842,801	229,509	216,165

III. SHARING DATA AND URBAN RHYTHMS

A. Rental Cohorts and Measurement Conventions

The two sharing sources record where and when rentals start and end in Bologna in 2023–2024. Bicycle records also distinguish electric and mechanical vehicles; 92.9% of retained rentals use electric bicycles. Recorded car kilometres are absent in 2024, so all distance comparisons use straight-line endpoint displacement, not route length. Rental duration is the interval between collection and return; for cars, it can include parking and intermediate stops.

Both cohorts require departure in the source year, both endpoints inside latitude 44.40–44.60 and longitude 11.20–11.50, and a finite duration of at least thirty seconds. Bicycle records additionally require a recognised bicycle type and at least 100 m displacement; plausible car returns near the origin remain included. Exact duplicates are removed before identifiers are discarded. The retained cohorts contain 4,574,629 bicycle rentals and 411,241 car rentals (Table I).

The distributions pool 2023–2024 rentals, with weekdays and weekends defined by departure calendar date. Bicycle durations use a valid-measurement subset: 0.5–180 minutes and endpoint-based speed at most 45 km/h. Car durations are uncapped. Counts and displacement use the full retained cohorts.

B. Temporal Patterns and Rental Characteristics

Figure 1(a,e) shows monthly activity from January 2023 to December 2024. Retained bicycle rentals rise from 1.73 million in 2023 to 2.84 million in 2024, and car rentals from about 195,000 to 216,000. The monthly curves also show an August dip in bicycle activity and a marked autumn rise for cars in 2024. These changes need not reflect changes in general travel: availability, service coverage and recording quality can all matter. In particular, October 2023 contains only one retained bicycle rental after filtering and must not be interpreted as a month without cycling. This coverage problem motivates excluding defective sharing periods from matched analyses.

Panels (b,f) separate the daily rhythm by departure day, pooling both years. Both services have a stronger morning component on weekdays and more overnight activity at weekends. For bicycles, 06–10 accounts for 16.4% of weekday departures but only 7.6% of weekend departures; the corresponding car shares are 14.0% and 9.5%. Conversely, the 00–06 share rises from 8.5% to 20.3% for bicycles and from 9.1% to 16.8% for cars. Afternoon and evening departures remain prominent in both services. Each curve is normalised within its service and day type, so these differences describe when rentals occur, not whether total weekend demand is higher. Pooling day types would conceal these different daily schedules.

In contrast to timing, typical endpoint displacement and duration differ little between weekdays and weekends within each service (Fig. 1(c,d,g,h)). Median bicycle displacement is about 1.64 km on weekdays and 1.67 km at weekends; car medians are about 2.59 km and 2.53 km. Later weekend activity is therefore not accompanied by systematically greater endpoint separation. Car rentals have larger median endpoint displacements, but these do not identify route length.

Median bicycle duration is about 10.3 minutes on weekdays and 10.8 at weekends; car medians are about 21.5 and 21.0 minutes. The distribution tails differ more substantially. Ninety-five percent of valid bicycle durations are within roughly 27.1 minutes on weekdays and 28.8 at weekends, whereas the corresponding car thresholds are about 8.1 and 15.5 hours. Similar weekday/weekend medians thus conceal a substantial difference in the car-duration tail. Long car intervals can include parking or stops and are not evidence of continuous driving.

C. Geographical Distribution of Sharing Activity

Figure 2 shows where the services are busiest. Rental departures and arrivals are pooled into common 500 m cells across four windows: 06–10, 10–15, 15–20 and 20–06. For arrivals A, departures D and valid observation hours H, activity is $(A + D)/H$. This measures endpoint turnover, not traffic along the streets between endpoints. Cells below 200 bicycle or fifty car endpoint events are left uncoloured, not interpreted as inactive.

Both services use matched observation windows, including continuous 20–06 nights assigned to the preceding evening. Every constituent hour must fall inside 2023–2024 and have usable bicycle-sharing coverage; clock-change dates are excluded for both sources. This leaves 697 common windows for each daytime bin and 693 nights. Matching valid hours avoids treating a recording defect as an absence of activity. The prediction experiments retain source-specific night conventions, distinct from these descriptive windows.

Both services concentrate activity in the central urban area, with additional concentrations in surrounding neighbourhoods. This shared urban focus does not imply equal coverage: bicycle rentals are far more numerous, and each service has its own colour scale. Colours permit comparisons across time windows within a service, but not of absolute volumes between services.

The centre remains active beyond the morning window. Both services reach their highest displayed cell-level activity rate in 15–20: about 48.5 bicycle and 2.81 car endpoint events per hour. These are maxima for individual cells, not citywide averages. The maps show sustained turnover, but intensity alone cannot distinguish areas with net arrivals from those with net departures. We examine this distinction through rental balances.

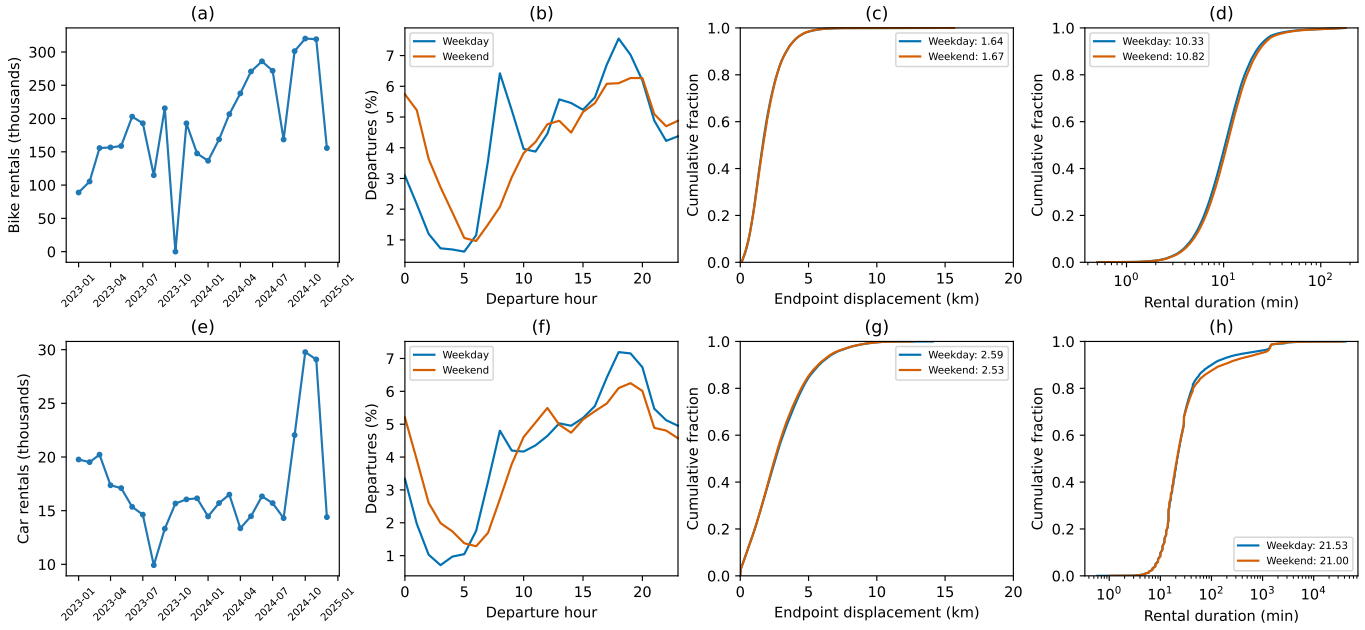


Fig. 1. Sharing volumes and distributions: bike-sharing in the first row (a–d), car-sharing in the second (e–h). (a,e) Monthly retained rentals in 2023–2024, in thousands. Other columns pool both years: (b,f) hourly departure shares, normalised within each service/day type; (c,g) cumulative straight-line endpoint displacement, not route distance (0–20 km shown, all observations in the denominator); (d,h) cumulative duration on logarithmic axes with panel-specific ranges. Bicycle durations use the valid-measurement subset; car intervals are uncapped and can include stops. Weekday is Monday–Friday and weekend Saturday–Sunday by departure calendar date. Curves are sampled empirical CDFs; legend medians use the full relevant cohorts.

TABLE II
REGIONAL RENTAL BALANCE, $100(A-D)/(A+D)$, IN THE MATCHED OBSERVATION WINDOWS. POSITIVE VALUES MEAN NET ARRIVALS. GEOMETRIC ZONES SUMMARISE THE MAPPED PATTERN, NOT INFERRED LAND USE. ALL RECORDED VIEWPORT CELLS ENTER THE SUMS.

Source	Area	06–10	15–20
Bike	Centre	+13.7	-2.3
Bike	Northern band	+0.9	-3.5
Bike	Remaining area	-32.5	+3.7
Car	Centre	+5.3	-4.8
Car	Northern band	+10.2	-3.8
Car	Remaining area	-27.7	+3.6

D. Net Balances and Daily Urban Polarities

Figure 3 shows the normalised balance, $(A-D)/(A+D)$, on a common scale for both services: red indicates net arrivals and blue net departures. A pale central cell can still be busy if arrivals and departures offset; a strongly coloured outer cell can have relatively few events. Balance describes the polarity of recorded movements, not their volume or the stock of available vehicles.

To summarise the maps, Table II uses three geometric zones. The centre comprises cell centres within 1.5 km of $(44.4949, 11.3426)$; the northern band comprises other cells at least 1.5 km north of that point; the remaining area contains all other cells within the map extent. Regional balances sum arrivals and departures before normalising, without applying the display threshold. Sign reversals and correlations instead use cells displayed in both relevant windows or services.

In 06–10, many outer cells show net departures, whereas the centre and selected northern and northwestern cells show net arrivals. Morning central balances are $+13.7\%$ for bicycles and $+5.3\%$ for cars; the remaining area has negative balances of -32.5% and -27.7%. The northern arrival surplus is clearer for cars ($+10.2\%$) than bicycles ($+0.9\%$), although local polarities can offset within this band. The pattern is compatible with movement towards central and northern activity areas. The Fiera and airport sectors provide geographical reference points for the northern and northwestern parts of the map, without identifying individual rental purposes.

In 10–15, the centre remains active but cell-level arrivals and departures become more balanced. Median absolute balance among displayed cells within the map extent falls from 0.34 to 0.08 for bicycles and from 0.31 to 0.05 for cars. A paler balance map therefore indicates weaker directional asymmetry, not absent movement. Cross-service balance correlation is weak and slightly negative at midday (-0.12): the same places can be busy without receiving and sending rentals in identical proportions.

In 15–20, central and northern regional balances turn negative and the remaining area positive (Table II), compatible with movement back towards outer destinations. High central turnover (Fig. 2) thus coexists with net outward movement. The morning-to-late-afternoon reversal is widespread: balance changes sign in 130 of 153 comparable bicycle cells (85.0%) and 166 of 225 car cells (73.8%). This describes a shared rhythm, not identified return journeys by the same people or direct OD flows between the zones.

The 20–06 window combines evening and early morning. Northern regional balances become positive again, at $+16.6\%$ for bicycles and $+7.0\%$ for cars, while the centre remains negative. Later hours therefore do not show a uniform outward

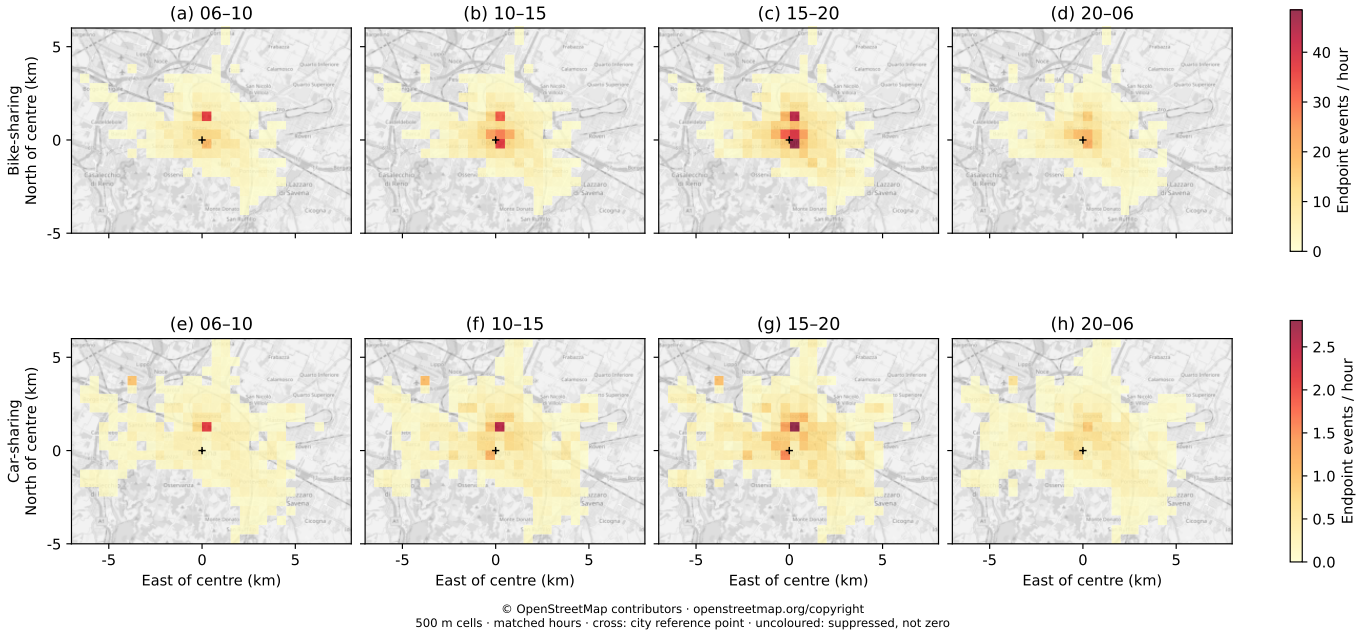


Fig. 2. Sharing intensity over a common Bologna background. Top: bike-sharing; bottom: car-sharing. Columns: 06–10, 10–15, 15–20 and continuous 20–06 nights. Colour is arrivals plus departures per valid hour in each 500 m cell. Each row uses one linear colour scale across its four windows; row scales differ because service volumes differ. Bicycle/car cells below 200/50 endpoint events are uncoloured, not zero. Both sources use matched valid hours. Coordinate labels are kilometres from $(44.4949, 11.3426)$, marked by a cross. Background: © OpenStreetMap contributors, https://www.openstreetmap.org/copyright; present-day context, not historical routes.

movement. Although observation hours are matched, this broad window cannot be interpreted as a single return-home period or used to identify a common travel purpose.

Cross-service balance correlation is 0.76 across 151 common displayed cells in the morning and 0.48 across 158 cells in late afternoon. Together with the temporal distributions, these results answer RQ1: the services reveal related urban polarities and weekday/weekend timing shifts. These patterns describe selective service activity. Whether sharing also helps estimate traffic at new locations or reconstruct counts at known locations remains a separate question, addressed below.

IV. Sensor Networks and Experimental Design

A. Reference Counts and Sensor Eligibility

Independent municipal sensors provide the reference counts. Bicycle counters record directional passages of all bicycles, not only shared bicycles [23]; road detectors count motor vehicles, not just shared cars or passenger cars [24]. These measurements count passages at specific street sections, not unique travellers or citywide journeys.

Figure 4 locates the experimental networks. The bicycle network contributes four locations: Ercolani, Parri, Sabotino and San Donato. These provide limited coverage around the central city, not observations in every neighbourhood. Paired devices and both directions are treated as one location when separating training and testing. Eligibility requires at least sixty usable dates in 2023 after consistency and complete-period checks. Six locations are excluded from the study cohort independently of prediction error. Conflicting duplicate records and contiguous all-zero device runs lasting at least 48 hours are excluded; every constituent device must be valid throughout a retained period.

The motor-vehicle network is denser, with 753 eligible devices in 2023 across 63 occupied square-kilometre blocks. Several instruments can occupy the same street area, so these are not 753 independent locations. Eligibility requires at least 180 usable dates in 2023 and station coordinates stable within 100 m. A usable period must have official recording completeness of 100% in every constituent hour [25]. Recording completeness does not guarantee correct physical measurement; the bicycle archive has no equivalent official completeness flag.

In 2024, 744 eligible motor-vehicle devices remain observed, including 736 with positive totals. Observations displaced more than 100 m from their training position are not evaluated as the same location. Such later displacements do not retroactively change the 2023 training cohort.

B. Prediction Targets and Validation Scenarios

Both experiments estimate bicycle or motor-vehicle passages during complete day or night periods when contemporaneous counts are unavailable. Models estimate mean hourly flow, which is multiplied by the observed period length; partial counter periods are not expanded. The scenarios differ in whether the target location's historical counts are available.

New locations (Scenario A, RQ2). Training uses 2023 only and excludes all counts at the test location. Bicycle tests leave out one entire location and train on the other three. For motor-vehicle spatial validation, links between devices within 100 m define 246 connected groups. Each group is assigned by its centre to a 1 km block, producing 63 occupied blocks in



Fig. 3. Normalised arrival–departure balance, $(A - D)/(A + D)$, using the same grids, matched hours, windows and suppression rules as Fig. 2. All panels share the $[-1, 1]$ scale: red indicates net arrivals, blue net departures. The map distinguishes the polarity of recorded movements from their intensity; it measures neither available vehicles nor street crossings. Background: © OpenStreetMap contributors, https://www.openstreetmap.org/copyright.

five fixed spatial folds (Fig. 4, right). Training also excludes devices within 250 m of a test device. Counts and predictions remain device-specific, rather than being summed into area totals.

Each fitted model is tested at excluded locations in 2023 and again in 2024 without refitting. The first test evaluates spatial transfer within the training year; the second evaluates whether it persists into the following year.

Known locations after a simulated sensor failure (Scenario B, RQ3). Training uses the 2023 counts of all eligible locations, and predictions concern the same locations in 2024. No spatial exclusion is applied because the target location's earlier measurements are available historical information. No 2024 counts enter prediction, including those from neighbouring counters, and no refitting or local calibration uses 2024 outcomes. This simulates sensor unavailability over the subsequent year, not the observed timing of hardware failures. Available 2024 counts are used only for evaluation, with exactly the same target observations as in Scenario A.

C. Input Sets and Historical Profiles

Four cumulative input sets test whether increasingly local sharing information adds value. Sharing activity and weather refer to the period being estimated, so this is reconstruction with contemporaneous inputs, not forecasting before those inputs are known:

1) *Context*: sensor position and distance from the centre, weekday, season, day/night, temperature and precipitation. Bicycle models also use direction; vehicle models include holidays. The known-location experiment additionally includes the local historical profiles described below.
2) *Citywide*: Context plus sharing departures per hour across the study area, describing the overall rhythm of service use. Bicycle models also include the electric-bicycle fraction.
3) *Endpoints*: Citywide plus sharing departures and returns per hour within 500 m of the sensor.
4) *Geometry*: Endpoints plus rates of chords passing within 100 and 250 m of the sensor, with the sensor's perpendicular projection inside the segment. Chords join rental origins and destinations; they are not reconstructed routes or verified sensor crossings.

Bicycle geometry distinguishes chords oriented towards and away from the centre; motor-vehicle geometry is unsigned. Returns are assigned to their own timestamps, while chords are assigned to departure time. A chord requires the completed rental's destination, so long car rentals can delay its availability. These inputs therefore support retrospective reconstruction, not a claim of real-time performance, and do not locate intermediate travel in time. Bicycle periods affected by a near-total source timestamp defect are excluded on the basis of sharing coverage, not prediction error.

At known locations, the pooled model also receives two frozen summaries of the sensor's 2023 history: the log mean hourly count by location, direction and day/night regime, and the same profile further separated by weekday. Means divide total counts by total observed hours; rates are floored at 0.01 before taking logs. A missing weekday profile falls back to its regime mean, and a missing regime to its location/direction mean, always using 2023 alone. Both Context and the sharing-enhanced models receive these profiles. Their comparison therefore measures what sharing adds beyond local history. No historical profile is supplied for a new location.

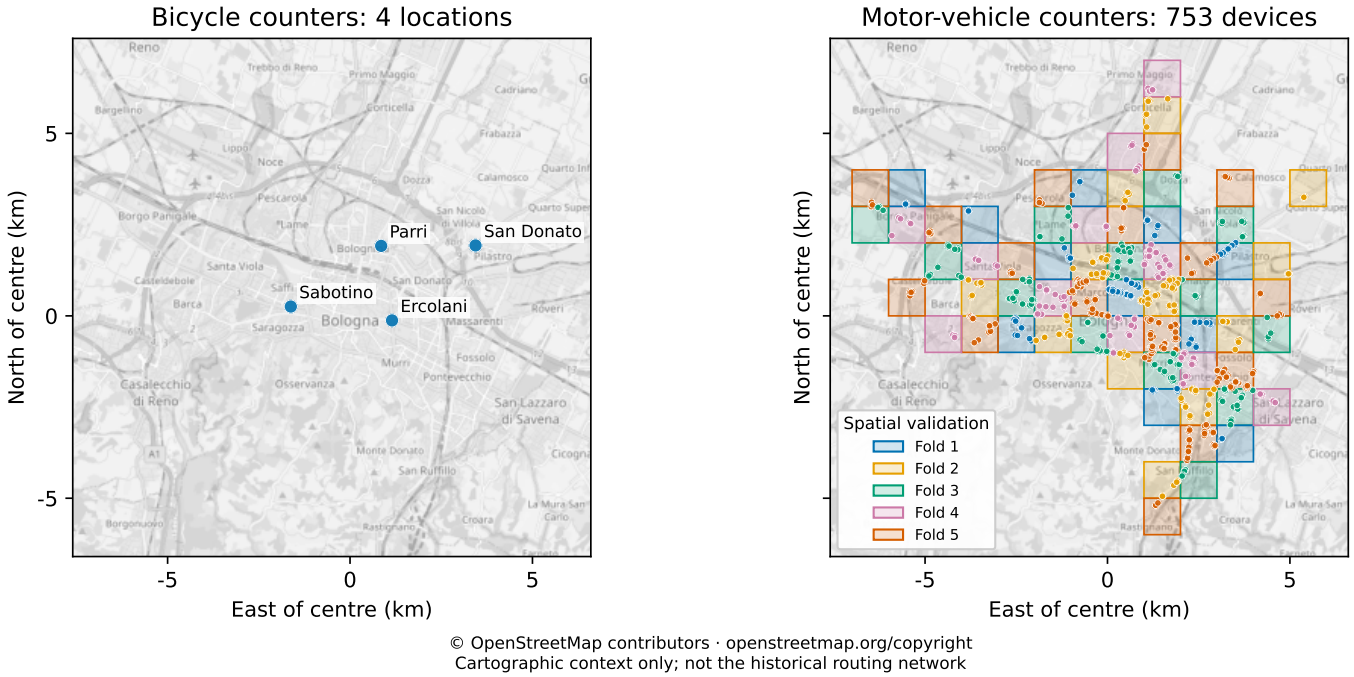


Fig. 4. Observation networks and spatial validation units, with the same geographical extent and scale. Left: four bicycle-counter locations. Right: 753 motor-vehicle devices (circles) and 63 occupied 1 km × 1 km cells, coloured by spatial validation fold. Each of the 246 proximity groups is assigned to the cell containing its centre; its devices share that cell's fold. The five folds are distributed sets of cells, not contiguous regions, and each is held out in turn. Counts and predictions remain device-specific; cells define validation and resampling units. Several devices overlap. The maps show the experimental networks, not every municipal sensor. Background: © OpenStreetMap contributors, https://www.openstreetmap.org/copyright.

D. Models and Reference Estimator

Each input set is tested with regularised Poisson regression, which provides a relatively simple count relationship, and gradient boosting with Poisson loss, which can capture nonlinear relationships. Bicycle and motor-vehicle models are fitted separately. Test observations are identical across input sets within each experiment, but absolute errors across the two domains do not constitute a controlled comparison between transport modes. Model settings remain fixed across input sets and scenarios, preprocessing uses training data only, and no tuning uses 2024 outcomes.

A separate reference, *uniform chord expansion*, multiplies the 250 m geometry proxy by a single factor calibrated on training counts. It tests whether sharing movements can simply be scaled up to represent all traffic and receives no local historical covariates in either scenario. Fitting details and parameter values are provided in the accompanying code.

E. Evaluation and Sensitivity Analysis

The main measure is pooled weighted absolute percentage error (WAPE): total absolute prediction error divided by total observed passages. Lower is better; a value of 40% means that absolute errors sum to 40% of the observed count. Busy locations carry more weight. We compare each sharing-enhanced model with the same learner using Context alone and report differences in percentage points; negative differences favour sharing. The known-location baseline includes historical profiles. Absolute error differences between scenarios also reflect training coverage and available history, not sharing alone.

Local evaluation checks whether pooled gains extend to individual locations. For new locations, we assess improvement in both test years at the four bicycle locations and 736 motor-vehicle devices with positive totals in both years; ties count as no improvement. Available dates need not be identical across years. For known locations, only 2024 is held out, so local improvement is reported for that year without a two-year persistence statistic. Zero-total devices contribute their absolute errors to pooled evaluation even though their individual WAPE is undefined. Bias, median and macro site errors provide complementary diagnostics.

Sensitivity checks use 2,000 paired resamples of bicycle locations or motor-vehicle spatial blocks, with predictions held fixed. These intervals do not establish citywide uncertainty or account for refitting; the bicycle analysis is further limited to four locations. Complete diagnostics, split membership and source hashes are retained in the reproducible outputs. The standalone pipeline rebuilds both experiments directly from the raw sources and verifies spatial exclusion, chronological separation, frozen history and equality of the 2024 evaluation targets.

V. Results and Discussion

Sharing's predictive value differs between the two scenarios. At new locations, gains are uneven and do not reliably persist into the following year. At known bicycle locations, sharing

improves gradient boosting beyond its history-aware baseline; motor-vehicle reconstruction remains strongest without sharing. Table III reports all configurations. We first examine pooled results in each scenario, then local benefits and the scope of the findings.

A. New Locations: Sharing beyond Spatial Context

For bicycle Poisson regression, Citywide reduces WAPE from 38.26% to 32.01% at excluded locations in 2023. The advantage reverses in 2024: Citywide reaches 40.61%, compared with 39.22% for Context alone. Thus citywide rental activity improves estimation within the training year but does not provide a dependable next-year adjustment at these locations.

Local sharing inputs do not consistently resolve this instability (Fig. 5). The Endpoints input set worsens bicycle Poisson estimates relative to Context in both years. Geometry improves bicycle boosting relative to its own Context baseline, including in 2024, but its 59.75% WAPE remains well above the 39.22% obtained by Poisson without sharing. A useful addition within one learner does not therefore identify the strongest estimator overall: model choice matters alongside sharing representation.

For motor traffic, Citywide gains are small in 2023 and disappear in 2024 for both learners. More spatial detail contributes little: Endpoints and Geometry leave boosting predictions exactly unchanged from Citywide. Under these settings, local sharing inputs do not improve estimation at new street sections.

Uniform chord expansion also falls short of the strongest contextual alternatives, and its error rises in the following-year test for both domains. The answer to RQ2 is therefore conditional: some sharing representations help particular models, but neither more endpoint detail nor a single expansion factor provides a reliably transferable estimate at new locations.

B. Known Locations: Reconstructing an Unavailable Sensor

At known bicycle locations, boosting with Geometry reduces WAPE from 17.68% for the history-aware Context baseline to 14.59%. Both configurations receive the same frozen local profiles, so the 3.09-percentage-point gain reflects sharing information beyond the historical traffic level and weekday/day-night pattern.

Most of the gain is obtained from citywide activity: WAPE falls to 15.34% before local endpoints or chords are added. The Endpoints input set provides a further improvement, whereas Geometry reduces WAPE by only another 0.14 percentage points, with a sensitivity interval spanning zero. This pattern provides stronger evidence for sharing as a supplementary temporal signal than for the specific value of chord geometry. The Geometry-minus-Context difference has a paired four-location interval of $[-5.06, -0.90]$ percentage points (Fig. 6).

The benefit does not extend to every model or domain. Bicycle Poisson deteriorates when sharing is added. For motor traffic, the strongest known-location configuration is Context boosting at 12.33% WAPE; sharing raises it to 12.86%. The 0.53-percentage-point deterioration remains positive throughout the block sensitivity interval, $[+0.46, +0.59]$. Endpoints and Geometry again leave motor-vehicle boosting predictions unchanged from Citywide, and Poisson also performs worse with sharing.

For RQ3, sharing therefore improves reconstruction at known bicycle locations with boosting, but not with Poisson or for motor traffic. Errors are much lower at known than at new locations even without sharing. This cross-scenario difference reflects available history and training coverage; sharing's additional contribution is established by comparisons within each scenario.

C. Local Benefit and Aggregate Accuracy

Pooled accuracy does not show whether sharing helps most locations or only a few influential ones. For bicycle Poisson with Citywide at new locations, three of the four locations improve in both years, although pooled WAPE is worse than Context in 2024. The loss at the remaining location outweighs the gains elsewhere. Pooled deterioration therefore does not imply that all local relationships have failed.

At known bicycle locations, all three sharing-enhanced input sets improve boosting at all four locations relative to Context. The pooled gain is therefore accompanied by a consistent direction of local improvement, rather than being driven by one location. This agreement between pooled and local results provides the strongest support for sharing in the tested known-location reconstruction setting.

Motor-vehicle gains are less widespread. With Citywide boosting at new locations, 271 of 736 common positive-total devices improve in both years (Fig. 7). At known locations, only 162 of 736 improve in 2024, alongside pooled deterioration. These local gains do not justify applying sharing enhancement indiscriminately across the network. For an operator considering whether to supplement a sensor's history, the distribution of gains matters as much as aggregate error.

D. What the Two Scenarios Establish

The experiments separate descriptive agreement from predictive usefulness. The related arrival and departure polarities in Section III do not establish accurate traffic estimation at new street sections. Conversely, limited spatial transfer does not rule out a useful sharing signal when local history is available, as the bicycle boosting result illustrates.

The gap between rental endpoints and street passages offers possible explanations for the uneven benefits. Through trips can have distant endpoints, nearby rentals can use parallel streets, and car-rental intervals do not locate intermediate driving in time. Local history supplies information about the monitored section that these service records lack. These are plausible mechanisms, not explanations established by the experiments; testing them would require verified routes and road attributes.

Several limitations define the scope of the evidence. The positive bicycle result concerns four selected locations and one subsequent year. The denser motor-vehicle network still contains spatially related devices and potentially unreliable measurements. Complete recording periods can contain almost no detected traffic, making individual WAPE values unstable;

TABLE III
POOLED WAPE (%). "NEW" EXCLUDES ALL COUNTS AT THE TEST LOCATION FROM TRAINING; "KNOWN" TRAINS ON ALL ELIGIBLE LOCATIONS IN 2023 AND RECONSTRUCTS COUNTS AT THOSE LOCATIONS IN 2024. WITHIN EACH DOMAIN, THE TWO 2024 SCENARIOS USE IDENTICAL TARGET OBSERVATIONS. LOWER IS BETTER.

Learner	Inputs	Bike new 23	Bike new 24	Bike known 24	Vehicle new 23	Vehicle new 24	Vehicle known 24
Boosting	Context	65.77	64.13	17.68	62.31	61.04	12.33
Boosting	Citywide	65.31	66.99	15.34	62.11	61.77	12.86
Boosting	Endpoints	73.01	73.63	14.73	62.11	61.77	12.86
Boosting	Geometry	59.82	59.75	14.59	62.11	61.77	12.86
Poisson	Context	38.26	39.22	26.00	64.85	63.63	14.73
Poisson	Citywide	32.01	40.61	35.78	64.59	68.90	23.66
Poisson	Endpoints	40.05	48.41	34.02	64.58	69.23	23.62
Poisson	Geometry	38.27	60.90	38.21	64.56	69.11	23.66
Scaling	Uniform chord	61.52	85.24	64.54	95.55	98.41	98.53

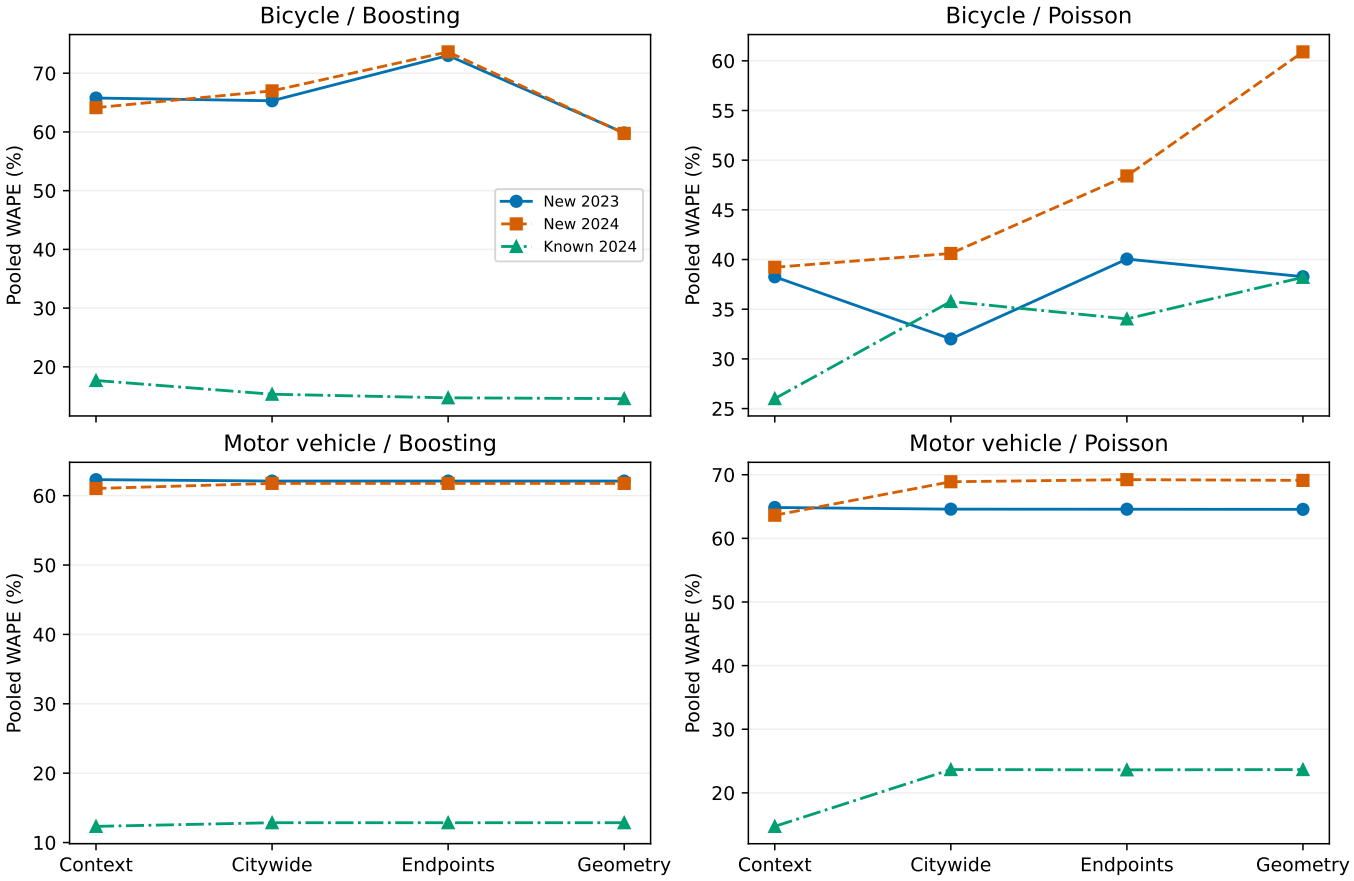


Fig. 5. Pooled WAPE across cumulative input sets, by target, learner and validation setting. The known-location Context baseline includes frozen 2023 local profiles. Panels use different scales; sharing's contribution is assessed relative to each curve's Context baseline.

pooled WAPE retains their absolute errors but cannot correct the observations.

Fixed model settings do not establish the best achievable estimator. The simulated full-year outage does not cover intermittent failures, and the exclusion of contemporaneous counts from the entire network is more restrictive than a single failed device with observed neighbours. These conditions also prevent interpreting differences between bicycle and motor-vehicle errors as an intrinsic mode effect.

In practice, sharing should be evaluated against the information available at the intended monitoring location. A new virtual counter requires demonstrated spatial transfer; reconstruction at a known location requires improvement beyond a credible historical baseline. Sharing meets the latter criterion for bicycle boosting in the tested setting, but this does not establish that it can generally replace sensors. Its supplementary value is scenario-dependent and should be assessed through both pooled accuracy and the distribution of local benefits.

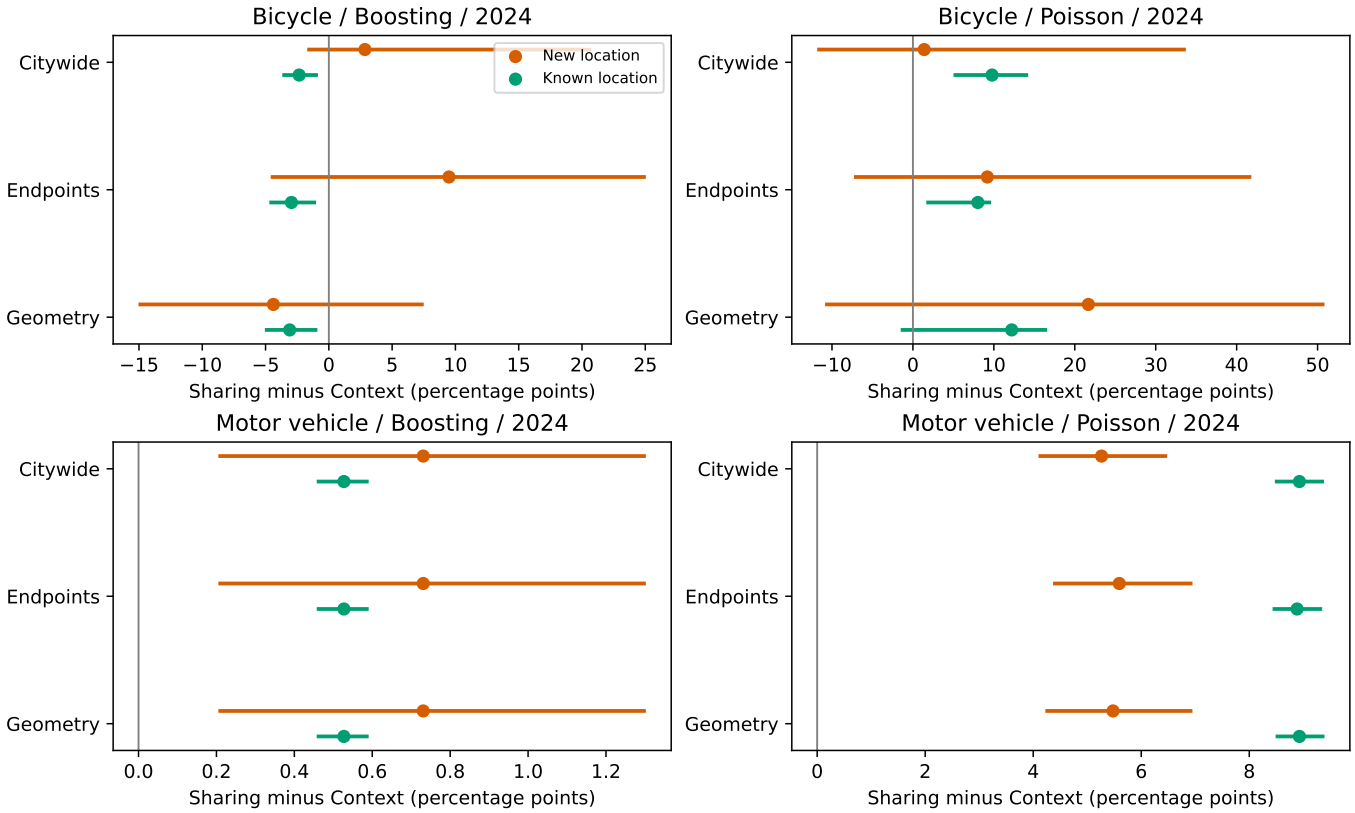


Fig. 6. Sharing-minus-Context WAPE differences in 2024 for the two scenarios, in percentage points. Negative differences favour sharing. Lines show paired spatial-unit resampling intervals with fitted predictions held fixed. Each contrast uses its scenario-specific baseline.

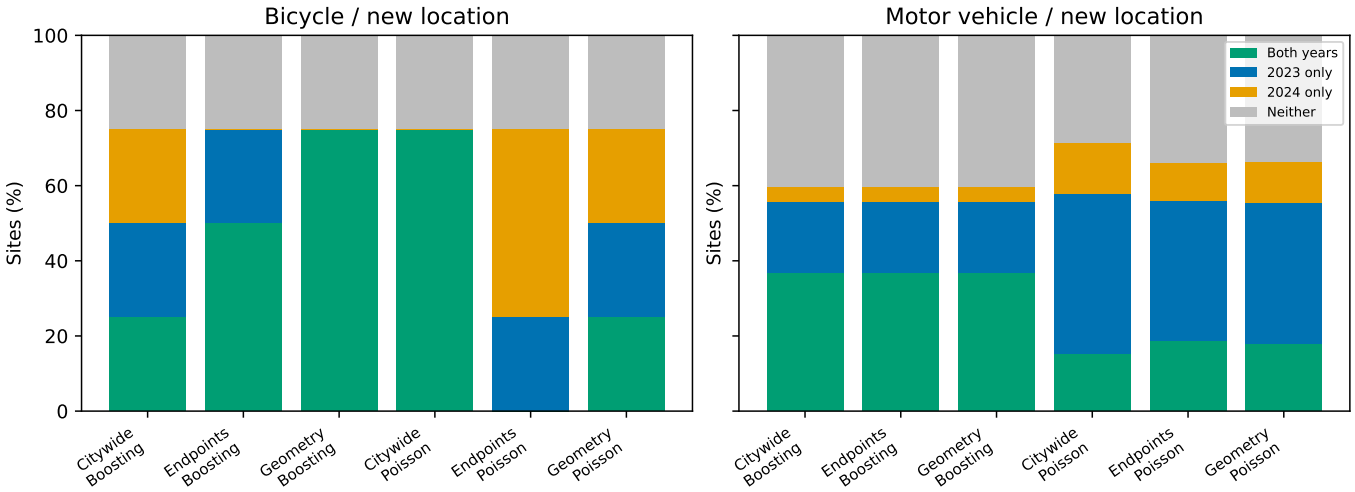


Fig. 7. Persistence of new-location improvements relative to each learner's Context baseline: improvement in both test years, one year only or neither. Denominators are four bicycle locations and 736 positive-total motor devices common to both years. Known-location reconstruction has only one test year and is not assigned a two-year persistence statistic.

VI. Conclusion

Sharing records reveal urban rhythms, but their traffic-monitoring contribution depends on the information available at the target location. For RQ1, the two services show related daily polarities and weekday/weekend timing shifts. For RQ2, pooled gains from citywide activity at excluded locations do not persist into 2024, and local sharing inputs do not consistently improve estimates. For RQ3, sharing reduces bicycle boosting WAPE from 17.68% to 14.59% at known locations, with improvement at all four locations. This benefit does not extend to bicycle Poisson or motor traffic, whose strongest known-location configuration is Context boosting without sharing.

Describing service activity, transferring count relationships

to new locations and reconstructing missing measurements at known locations are distinct tasks. Sharing's incremental contribution must be demonstrated separately for the latter two, using credible scenario-specific baselines and both pooled and local diagnostics. Evidence from one missing-count scenario cannot substitute for evaluation in the other.

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